RESEARCH ARTICLE · SPECIAL FUNCTIONS

The Tahawus Function: An Extension of the Lambert W Function

The inverse of the map $x \mapsto (x^x - 1)/x$, the constant named for a ghost town, and its arithmetic

Justo Tapiador García

Universidad de Alicante (UA)

An expanded and rigorous version of the post Tahawus: An Extension of the Lambert W Function
"Hola, os presento a Tahawus: una extensión de la función W de Lambert" (3 April 2018)

Tahawus function · Lambert W · tetration · super-quadratic · Gelfond–Schneider · transcendence · power tower

2020 Mathematics Subject Classification: 26A18 · 33E30 · 11J81

September 27, 2026